

that our classifier with about 3 million parameters has been added as a fully connected layer to the pre-trained model.

Now we will freeze the `conv_base` and train only our own. Follow these steps:

1. Set the trainable property of the `conv_base` layer to `False` since we do not want to train it.
2. Now, print the number of weights after freezing the `conv_base`. It is down to four after freezing from 492 before freezing. This verifies that the pre-trained model has been freeze.
3. Finally, we to fine-tune some of the hyperparameters. For instance, increase the learning rate slightly from 0.0001 in the last model to 0.0002(2e-5) and compile the model.
4. Use the same plot code to plot the various graphs of training and validation accuracy with the number of epochs.

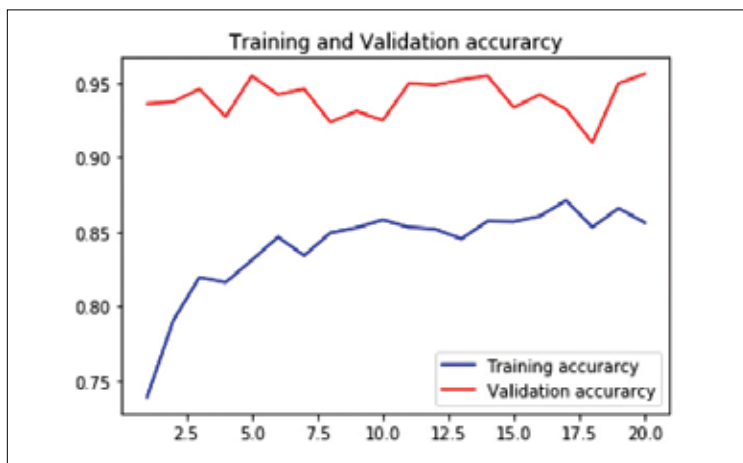


Figure : Training and validation accuracy of the model using transfer learning

Looking at the graph, we find that the validation accuracy decreases on increasing the number of the epoch. Hence, we have fine-tuned the number of the epoch from 64 in the previous model to 20 in this model. Finally, we achieved an accuracy of about 96% in just 20 epochs.

Finally, we can see clearly that transfer learning is a very efficient approach to increase the accuracy of our model. However, we also found that with the increasing number of training parameters it is not consistent